

Shuwei ZHAO

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Education

The Chinese University of Hong Kong

Bachelor of Engineering in Electronic Engineering

Sep. 2023 – May 2027

Hong Kong SAR

– CGPA: **3.911/4.000**; Major GPA: **3.967/4.000**.

– **Engineering Elite Stream** (2023–2026), a selective academic enrichment programme for high-performing engineering students.

ETH Zurich (Swiss Federal Institute of Technology Zurich)

Exchange Student in Electrical Engineering and Information Technology

Feb. 2026 – Aug. 2026

Zurich, Switzerland

Publications

- [1] **Shuwei Zhao**[†], Ruizhou Zhao[†], Zhiqing Tang, Wenchao Yue, and Hongliang Ren. “A Magnetically Steerable Micro-Camera Catheter for Bronchial-tree Phantom Navigation.” *Accepted, International Conference on Biomimetic Intelligence and Robotics (ICBIR)*, 2026.
- [2] Ruizhou Zhao[†], Yichen Chu[†], **Shuwei Zhao**, Wenchao Yue, Raymond Shing-Yan Tang, and Hongliang Ren. “Bioinspired Kirigami Capsule Robot for Minimally Invasive Gastrointestinal Biopsy.” *Accepted, IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [arXiv:2602.06207](https://arxiv.org/abs/2602.06207).

[†]Equal contribution.

Research Experience

Magnetically Steerable Catheter System for Industrial Inspection

Jul. 2026 – Aug. 2026

Research Assistant | Xi'an Jiaotong University

Advisor: Prof. Qiji Ze

- Designed a motorized translating-pulley catheter accumulator to manage excess catheter length during insertion and retraction, reducing uncontrolled bending and entanglement without winding the catheter onto a conventional reel; established the geometric relationship $\Delta L \approx 2\Delta x$.
- Developed the mechanical assembly in Fusion 360 and SolidWorks, integrating a 120 mm guide pulley, 608 bearings, an MGN12H linear carriage, and a T8 lead-screw stage with custom 3D-printed adapters and support components.
- Implemented the complete motion-control stack using a Python/Tkinter interface, USB serial communication, Arduino UNO firmware, and a TB6600 stepper driver, supporting manual calibration, jogging, fixed-distance motion, adjustable speed and acceleration, software travel limits, and communication-watchdog protection.

A Magnetically Steerable Micro-Camera Catheter for Bronchial-tree Phantom Navigation

Jun. 2026 – Jul. 2026

Research Assistant | The Chinese University of Hong Kong

Advisor: Prof. Hongliang Ren

- Designed and fabricated a millimeter-scale catheter integrating an LED micro-camera, heat-shrink encapsulation, a 0.2-mm nitinol support wire, and a wrapped hard-magnetic Ecoflex actuation segment.
- Built an external permanent-magnet steering setup and characterized insertion force under dry and wet conditions together with distal deflection in horizontal and vertical planes.
- Demonstrated magnetically assisted steering through primary and secondary bronchial bifurcations while obtaining forward-looking intraluminal images; led manuscript preparation as co-first author.

Soft Miniaturized Intraductal Imaging Catheter

Mar. 2026 – Jun. 2026

Research Assistant | ETH Zurich, Multi-Scale Robotics Lab (MSRL)

Advisor: Prof. Bradley Nelson

- Designed and fabricated a soft, sub-millimeter imaging catheter using 0.5-mm and 0.7-mm nitinol supports, balancing flexibility, pushability, insertion resistance, and distal miniaturization through external illumination.
- Built a four-load-cell force-measurement platform with Arduino/Python data acquisition; completed ten insertions per condition and compared the prototypes with a 1.1-mm rigid reference.
- Identified the 0.5-mm nitinol configuration with silicone coverage as the lowest-force design and validated real-time imaging in a 3D-printed silicone breast-duct phantom; semester project grade: **5.5/6**.

Polymer Material-Based Jumping Robot

Sep. 2025 – Present

Research Assistant | The Chinese University of Hong Kong

Advisor: Prof. Hongliang Ren

- Designed and fabricated an Arduino-controlled pneumatic jumping robot using a double-layer spherical-cap silicone actuator, achieving repeatable jumping in benchtop experiments.
- Derived geometric models for layer-junction coordinates and optimized the actuator radius and opening angle for deformation and elastic-energy storage.
- Developed air-pump control and pressure-release logic and evaluated material stiffness and inflation-release cycles against jump height, frequency, and repeatability.

Bioinspired Kirigami Capsule Robot for Minimally Invasive Gastrointestinal Biopsy

Jun. 2025 – Aug. 2025

Research Assistant | The Chinese University of Hong Kong

Advisor: Prof. Hongliang Ren

- Integrated A4988-based motor-control electronics and developed Arduino pulse-control software for a 17-mm biopsy capsule; built an angular calibration platform yielding approximately 18.1° per pulse and $R^2 = 0.99$.

- Conducted ex vivo biopsy experiments on porcine gastric and intestinal tissues, measuring tissue penetration, interaction forces, and retrieved sample mass.
- Analyzed experimental data and authored validation content for the accepted ICRA paper; experiments demonstrated shallow penetration (median ~ 0.61 mm) and repeatable tissue collection.

Autonomous Tracking and Grasping Robot

Oct. 2024 – Jan. 2025

Course Project | The Chinese University of Hong Kong

Advisor: Ms. Lai Yin NGAN

- Built an autonomous mobile robot for line tracking, object acquisition, transportation, and placement within a four-minute competition task.
- Integrated a multi-sensor tracking scheme with a stepper-motor-driven gripper and high-friction contact surfaces for stable object manipulation.
- Optimized inertial compensation, action sequences, and state-transition logic in C++ to improve tracking stability and task efficiency.

Honors & Scholarships

Song Qingling Scholarship Zhiyuan Education Fund; RMB 320,000	2023–2027
Dean’s List Faculty of Engineering, top 10%	2023, 2024
United College Head’s List Outstanding academic performance	2023, 2024
Charles Kao Scholarship	2024, 2025
Engineering Elite Stream Scholarship	2023, 2024
Arthur and Louise May Memorial Scholarship	2023–2025
Second Prize, Shandong Provincial Mathematics Competition	2022

Technical Skills

- Mechanical Design & CAD:** SolidWorks (independent part and assembly design), Fusion 360, Onshape
- Embedded Systems & Electronics:** Arduino, motor control, sensor integration, schematic and PCB design, KiCad, EasyEDA Pro (JLCEDA)
- Programming:** C/C++, Python, MATLAB, LaTeX
- Experimental & Fabrication:** 3D printing, silicone/Ecoflex molding, nitinol catheter fabrication, force sensing and load-cell calibration, magnetic actuation, optical/infrared sensing, ex vivo tissue experiments